Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE: solaress1
Tax ID 0575566002051
www.solaress.org · www.โซลาร์เอสส์.com

Example report from the Solar ESS Free System Check, published with the customer's details removed. All figures are real system data.

Plant Performance Report · Rev 03 · 11 November 2026

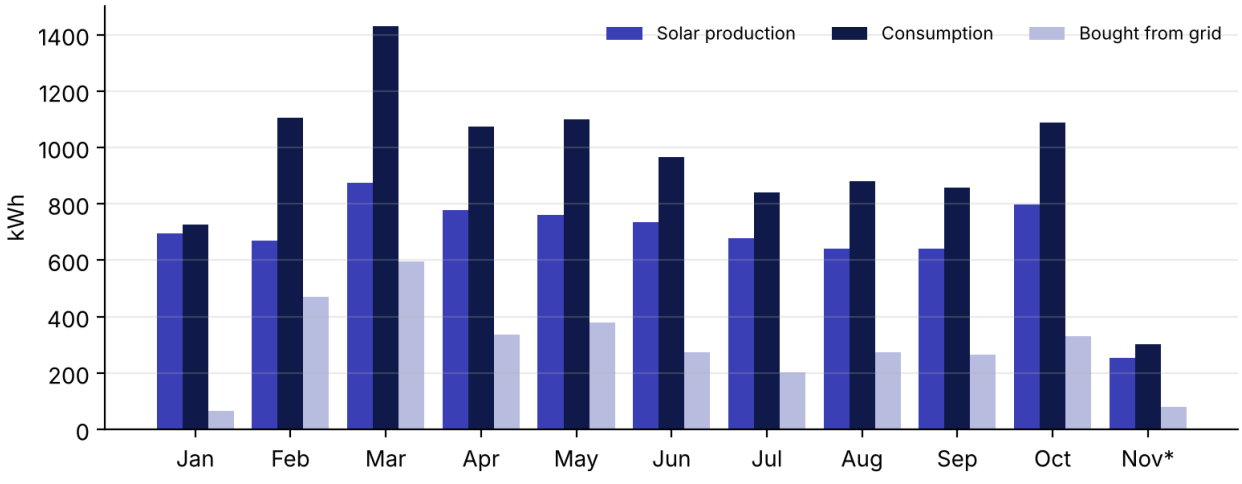
Example report — 6.48 kWp Hybrid Solar + 16 kWh Battery

Home in eastern Bangkok · Deye 5 kW hybrid inverter · 12 × LONGi 540 W panels · battery backup, zero export · in operation since April 2025 · data from DeyeCloud up to 10 Nov 2026 14:32

Total production	2026 production	Self-sufficiency 2026	Grid purchase 2026
12.95 MWh	**7,522 kWh**	**68.5%**	**3,266 kWh**
since Apr 2025	≈3.7 kWh/kWp/day	solar + battery	31.5% of use

Summary. The PV array is performing well with no sign of degradation or soiling. The main issue is the battery state-of-charge (SOC) reading, which drops to 0% overnight while usable energy remains, so the home switches to grid power earlier than necessary. On most sunny days the battery does not reach full charge, so additional PV capacity would reduce grid purchases more than additional battery storage.

Monthly energy 2026



* November = 1–10 Nov 2026.

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Month	Production (kWh)	Consumption (kWh)	Grid purchase (kWh)	Self-sufficiency
Jan	694.2	726.8	65.1	91%
Feb	669.5	1,105.1	470.0	57%
Mar	874.5	1,429.7	594.2	58%
Apr	776.7	1,073.5	336.3	69%
May	760.9	1,098.7	378.8	66%
Jun	735.9	965.3	273.7	72%
Jul	677.2	838.9	202.0	76%
Aug	640.3	880.5	272.6	69%
Sep	641.2	857.3	263.9	69%
Oct	798.2	1,087.9	329.9	70%
Nov*	253.7	302.7	79.2	74%
Total 2026	**7,522.3**	**10,366.4**	**3,265.7**	**68.5%**

Energy flows 2026

Where the solar energy went	kWh	Share
Used directly by the home	3,977	52.9%
Charged into the battery	3,540	47.1%
Exported to grid (zero-export setting)	5	0.1%

Where the home's energy came from	kWh	Share
Solar directly	3,977	38.4%
Battery discharge	3,124	30.1%
Grid purchase	3,266	31.5%

Battery round-trip: 3,540 kWh charged, 3,124 kWh discharged (≈88%), normal for an LFP system.

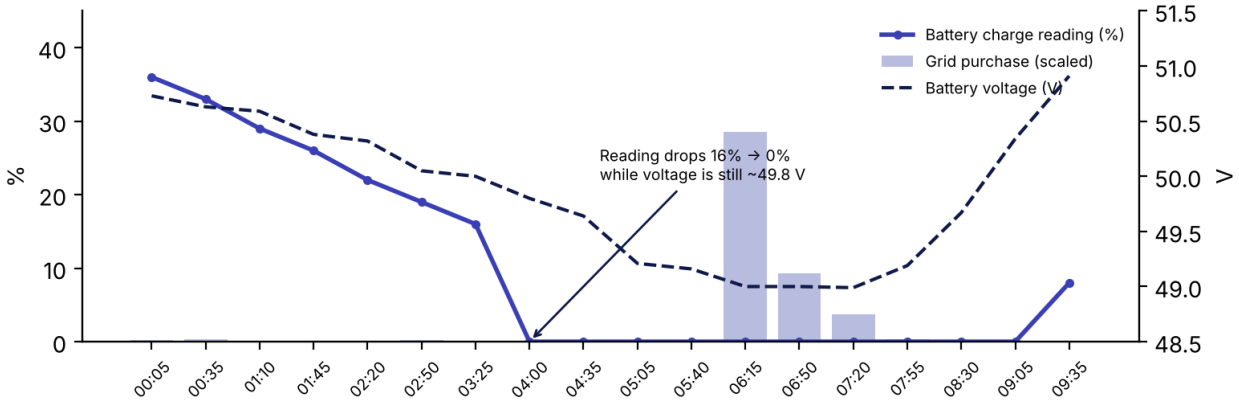
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Findings

GOOD **PV production is healthy.** 2026 averages ≈727 kWh/month (≈3.7 kWh/kWp/day on the 6.48 kWp array), in line with the typical Bangkok yield. Clear-day output in November is ≈25–29 kWh/day with a peak of 5.28 kW; the seasonal dip in Jul–Sep (≈640 kWh) is normal for the rainy season. The array is 12 × LONGi 540 W panels (6.48 kWp DC) on a Deye 5 kW hybrid inverter, a DC/AC ratio of 1.3.

ATTENTION **Battery SOC reading is inaccurate.** Overnight the reading drops from 16% straight to 0% (≈03:40 on 10 Nov) while the battery voltage is still ≈49.8 V, around 15–20% remaining for an LFP battery. The reading then stays at 0% until ≈09:00 and the home buys 1.1–1.6 kW from the grid between ≈02:00 and 07:00. Battery capacity is set at 16 kWh while DeyeCloud estimates 17.25 kWh.



Night of 9–10 Nov 2026: battery charge reading, battery voltage and grid purchase.

OPPORTUNITY **The system is energy-limited, not storage-limited.** On typical sunny days the battery does not fill (max 67% on 5 Nov, 76% on 15 Mar, 49% on 15 Jun) and the home still buys 8–13 kWh/day. Night load cycles between ≈0.5 and 1.5 kW (likely air-conditioning).

OPPORTUNITY **Occasional PV curtailment.** On low-use days (e.g. 15 Oct, 13.9 kWh) the battery is full by ≈11:00 and, with zero export, PV output is held back to 300–600 W for the afternoon.

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October 2026 vs October 2025

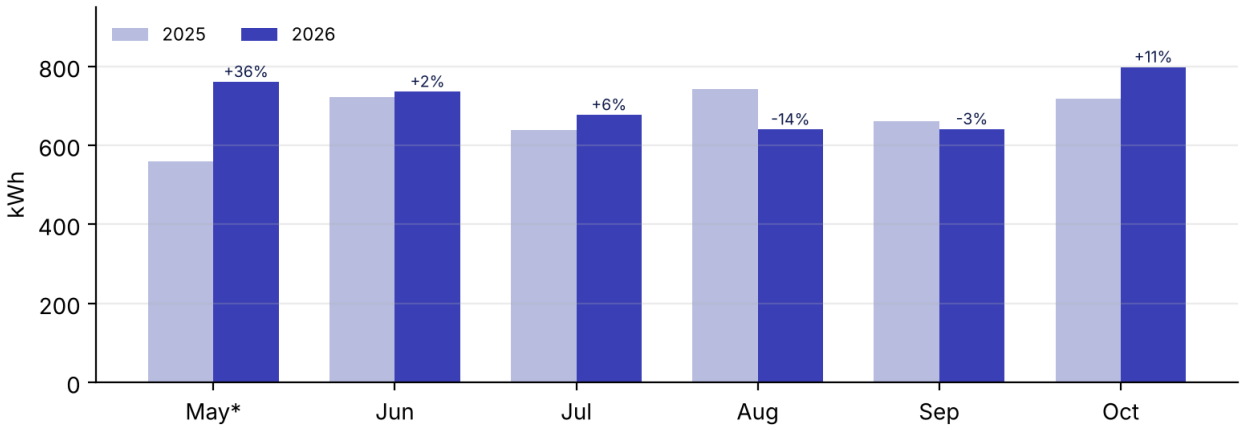
	Oct 2025	Oct 2026	Change
Solar production (kWh)	719.0	798.2	+11%
Daily average production (kWh/day)	23.2	25.7	+2.5
Home consumption (kWh)	903.5	1,087.9	+20%
Grid purchase (kWh)	228.3	329.9	+45%
Battery charged (kWh)	380.0	367.1	–3%
Battery discharged (kWh)	337.3	327.2	–3%
Battery round-trip efficiency	88.8%	89.1%	—
Self-sufficiency	75%	70%	–5 pts

- **Production +11%**, mainly weather: Oct 2025 had six cloudy days (13.6–19.8 kWh) against two in Oct 2026; best clear days reach ≈31 kWh in both years, so there is no sign of panel degradation.
- **Consumption +20%**: typical days rose from ≈24–37 kWh to ≈33–44 kWh, which is why grid purchase rose 45% despite higher production.
- **Battery unchanged**: ≈10.6–10.9 kWh delivered per night in both years (about two-thirds of 16 kWh), consistent with the SOC reading issue; efficiency stable at ≈89%.

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Month-by-month, May–October 2025 vs 2026



Month	Production (kWh)	Consumption	Grid purchase	Battery discharge	Self-sufficiency
May*	560 → 761 (+36%)	+16%	–6%	+41%	57% → 66%
Jun	721 → 736 (+2%)	+1%	0%	+6%	71% → 72%
Jul	639 → 677 (+6%)	–11%	–40%	+13%	64% → 76%
Aug	743 → 640 (–14%)	–1%	+43%	–28%	79% → 69%
Sep	661 → 641 (–3%)	+3%	+23%	–17%	74% → 69%
Oct	719 → 798 (+11%)	+20%	+45%	–3%	75% → 70%
May–Oct total	**4,043 → 4,254 (+5%)**	**+5%**	**+4%**	**–1%**	**70% → 70%**

* The system started operating on 25 April 2025; May 2025 was likely a settling-in month and is not a fair comparison.

- **Production is stable to slightly higher** (+5% over May–Oct). The only clear drop, August (–14%), matches a –28% drop in battery charging, i.e. weather.
- **Grid purchase follows consumption, not solar:** months with higher use (Aug–Oct 2026) bought more from the grid.
- **The battery cycles the same as last year** (1,844 → 1,819 kWh over May–Oct) with no loss of efficiency.
- **November so far:** 25.4 kWh/day in 2026 vs 20.5 kWh/day in Nov 2025.

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Recommendations

#	Action	Benefit
1	Check battery–inverter BMS communication (Lithium mode), the battery capacity setting and a full 100% charge to recalibrate SOC.	Use the full battery capacity at night; less early-morning grid purchase.
2	Plant record updated: 12 × LONGi 540 W = 6.48 kWp DC on a Deye 5 kW hybrid inverter.	Accurate monitoring and yield tracking.
3	Add ≈1.5 kWp PV only if the inverter is an SM2 model (8 kW PV input); older models take max 6.5 kW, already reached (check the label on the side of the inverter).	Battery fills more often; lower grid purchase.
4	Run large loads (water heater, pump, laundry, EV charging) around midday.	Uses solar that would otherwise be curtailed.

Get a free check of your own system at www.solaress.org/system-check, or contact us at 063 151 6694 / 089 632 5829 or LINE: solaress1.

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